SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I RAHIMA BANU N-192525009 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

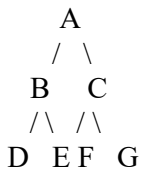
Signature of the student:
RAHIMA BANU N

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



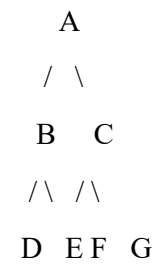
Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.
Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

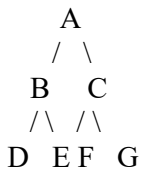
Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

SOLUTIONS:

Question 1: Breadth-First Search (BFS)

Given graph:



Algorithm: BFS visits nodes level by level using a Queue (FIFO).

Dry Run Table

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

1. BFS Traversal Order

A → B → C → D → E → F → G

2. Queue contents

- Initially: [A]
- After A: [B,C]
- After B: [C,D,E]
- After C: [D,E,F,G]
- After D: [E,F,G]
- After E: [F,G]
- After F: [G]
- After G: []

Question 2: Depth-First Search (DFS)

Algorithm: DFS explores one branch completely before backtracking. It uses a Stack (LIFO).

Dry Run Table

Iteration	Visited Node	Stack (top at right)
1	A	C, B
2	B	C, E, D
3	D	C, E
4	E	C
5	C	G, F
6	F	G
7	G	Empty

DFS Traversal Order

A → B → D → E → C → F → G

Explanation:

1. Start at A.
2. Visit B.
3. Visit D.
4. Backtrack and visit E.
5. Backtrack to A and visit C.
6. Visit F and G.

Question 3: 8-Puzzle using BFS

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Algorithm: BFS explores all possible states level by level until the goal is found.

For the blank tile, use the move order Up, Down, Left, Right. Avoid adding duplicate states.

Dry Run Table

State notation: 136/502/478 represents three rows.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	136/502/478	Initial	4 states
2	106/532/478	4 states	5 states
3	136/572/408	5 states	6 states
4	136/052/478	6 states	7 states
5	136/520/478	7 states	8 states

The first five expansions are shown here; BFS continues until it reaches the goal.

Complete shortest solution path

Step	Blank Move	State
0	Initial	136/502/478
1	Right	136/520/478
2	Up	130/526/478
3	Left	103/526/478
4	Down	123/506/478
5	Left	123/056/478
6	Down	123/456/078
7	Right	123/456/708
8	Right	123/456/780

Goal reached in 8 moves

Answers to the questions

1. Which node is expanded in each iteration?

The first five expanded states are:

1. 136/502/478
2. 106/532/478
3. 136/572/408
4. 136/052/478
5. 136/520/478

2. How many states are generated in total?

The number of states depends on the duplicate-checking and move-generation procedure. The complete count is not provided in the question paper, so it should be obtained by tracing the full BFS.

3. Complete solution path

Initial state → Right → Up → Left → Down → Left → Down → Right → Right → Goal state.

4. Why does BFS always find the shortest solution?

BFS explores states level by level. Therefore, the first goal state found has the minimum number of moves in an unweighted problem.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
Criteria	**Weightage (%)**	**Level 4 – Exemplary (4)**	**Level 3 – Proficient (3)**	**Level 2 – Developing (2)**	**Level 1 – Beginning (1)**
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.